

Genetic report for longevity

Hormones

Testosterone and SHBG, estradiol and aromatase, 5-alpha-reductase, thyroid, cortisol and stress axis, GH/IGF-1 axis and prolactin

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Hormones bind metabolism, muscle, bone, mood and aging speed. This report reads the genes of testosterone and SHBG, estradiol and aromatase (CYP19A1), 5-alpha-reductase (SRD5A2), thyroid (DIO1, DIO2, TSH and hypothyroidism scores), cortisol and stress axis (FKBP5, NR3C1, SERPINA6), GH/IGF-1 axis and prolactin.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

27

variants

21

in the chip

4

imputed
($DR2 \geq 0.8$)

2

not found

11

Scores with
percentile

Overall reading. Hormonal genetics is balanced, with some favourable points and almost no unfavourable. In testosterone, the men's score is at the 58th percentile and SHBG in the 66 (glucose protection side), with 1 allele 356N of SHBG that extends to SHBG and without the rare variant P156L; the genetic DHEA-S is high (percentile 11 in the DHEA-S low). The genetic estradiol is on the low side (the 13th percentile), despite 2 copies of the allele of CYP19A1 that rises; in man, estradiol is what protects the bone, so it should not be further downloaded. Thyroid: DIO2 Thr92/Thr92 (normal conversion of T4 into T3 in tissues), TSH genetics on the low side (the 19th percentile), T4 free typical (28) and hypo and hyperthyroidism in the medium (42 and 56); FOXE1 with 1 risk allele (small effect). In stress, FKBP5 is CC (without the T allele that prolongs cortisol response) and genetic cortisol is typical (the 33rd percentile). In the GH/IGF-1 axis, the genetic IGF-1 is at the 34th percentile (lightly low, the preferred side for longevity) and the IGF1R is heterozygous; the genetic prolactin is low (the 12th percentile).

Favourable

- **Prolactin (GWAS loci):** the 12th percentile
- **DHEA-S (loci of GWAS):** the 11th percentile
- **SHBG rs6259 AG:** D356N: allele A (356N) prolongs the half-life of SHBG and increases it; G = common.
- **SHBG rs6258 CC:** P156L: T allele reduces the binding of SHBG to testosterone (low SHBG); rare.
- **DIO2 rs225014 TT:** Thr92Ala: C allele (92Ala) lowers the activity of DIO2 and T3 in tissues (brain, muscle); in treated hypothyroidism, some Ala/Ala carriers feel better with associated T3.
- **FKBP5 rs1360780 CC:** FKBP5 (2 intron): T allele increases the induction of FKBP5 by cortisol, makes the glucocorticoid receptor less sensitive and prolongs the stress response; interacts with stress in childhood (depression, PTSD).

Unfavourable

None.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 21 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Testosterone and SHBG

SHBG transports testosterone and estradiol; its genes (SHBG, and liver loci) explain much of the total testosterone. Human testosterone has its own loci (JMJD1C, FAM9B in X). Genetically higher SHBG protects from 2 type diabetes in mendelian randomization.

Ageing mechanisms: Endocrine dysregulation; Sarcopenia; Thin Mass Loss; Insulin resistance

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Testosterone in man (3985 variants)	No judgement (informative)	the 58th percentile · z=0.18	Higher total testosterone score. Without judgment: it is not always better (prostate, hematocrit). PGS001988 (PGS Catalog; Penalized regression; Testosterone (male only); PGP000263, PMID 34995502). Variants with genotype: 3857 of 3985 (covering of 96.8% coefficients).
SHBG (19 328 variants)	Typical / intermediate	the 66th percentile · z=0.39	Higher score = higher SHBG (protects from diabetes type 2; low free testosterone). PGS000694 (PGS Catalog; penalized regression; SHBG [nmol/L]; PGP000128, PMID 33462484). Variants with genotype: 17214 of 19328 (covering of 88.4% coefficients).
DHEA-S (loci of GWAS)	Favourable	the 11th percentile · z=-0.95	Higher score = lower DHEA-S. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 18 of 18 loci with genotype. Studies: Miranda JP (DHEAS levels); Pott J (DHEAS levels); Shin SY (DHEA-S-to-4-androsten-3beta,17beta-diol disulfate 2 ratio); Surendran P (Dehydroisoandrosterone sulfate (dhea-s) levels); Zhai G (Dehydroepiandrosterone sulphur levels). Variants with genotype: 18 of 18 (covering of 100.0% coefficients).

Favourable

- DHEA-S (GWAS loci) (the 11th percentile)
- SHBG rs6259 AG
- SHBG rs6258 CC

Unfavourable

None.

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SHBG	rs6259 17:7633209 · G/A/C	AG	imputation DR2=0.98	G (1; 90%)	moderate	D356N: allele A (356N) prolongs the half-life of SHBG and increases it; G = common.
SHBG	rs6258 17:7631360 · C/A/T	CC	chip	T (0; 1%)	strong	P156L: T allele reduces the binding of SHBG to testosterone (low SHBG); rare.

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SHBG	rs1799941 17:7630105 · G/A	AG	chip	G (1; 74%)	strong	SHBG promoter: allele A increases SHBG (~10%); G = lower SHBG.
FAM9B	rs5978298 X:9135052 · A/G/T	A	imputation DR2=0.90	T (0; 46%)	moderate	FAM9B (X): human testosterone locus. Meaning: GWAS Catalog: Total testosterone levels (2021; p=4e-08); T ↑ allele

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
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JMJD1C	rs10822184 10:63577393 · T/C/G	TT	chip	—	moderate	JMJD1C: total testosterone locus in man.
AR	rs6152 X:67545785 · G/A/T	G	chip	—	moderate	Androgen receptor (X): allele A is associated with less baldness; the length of repetition CAG (not read here) is the one that most changes sensitivity to androgens. Information.

Estradiol, aromatase and 5-alpha-reductase

In humans, estradiol comes from the aromatization of testosterone (CYP19A1) and is essential for bone and vascular function. 5-alpha-reductase type 2 (SRD5A2) converts testosterone into DHT (prostate, hair).

Ageing mechanisms: Decline of the Estrogen Signal; Endocrine dysregulation; Osteoporosis

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Estradiol (9583 variants)	No judgement (informative)	the 13th percentile · z=-1.16	Higher score = estradiol. No judgment. PGS002916 (PGS Catalog; statistical method; Estradiol; PGP000393, PMID 36152628). Variants with genotype: 4803 of 9583 (covering of 30.4% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP19A1	rs2470152 15:51302775 · G/A/C	GG	imputation DR2=1.00	—	moderate	CYP19A1 (aromatase): locus of estradiol in man.
CYP19A1	rs727479 15:51242350 · C/A/G/T	AA	chip	A (2; 63%)	moderate	CYP19A1 (4 intron): associated with estradiol and estradiol/testosterone ratio. It has 2 copies of the allele that rises to estradiol (without judgment: in man, estradiol protects the bone). Meaning: GWAS Catalog: Estradiol levels (Eriksson AL 2018; p=8e-30); Allele A ↑
SRD5A2	rs523349 2:31580636 · G/A/C/T	CC	chip	—	weak	V89L: the allele L (C in strand +) slightly lowers the activity of 5-alpha-reductase. Information.
CYP17A1	rs743572 10:102837395 · A/G/T	AG	chip	—	weak	CYP17A1 -34T>C: inconsistent associations with sex hormones. Information.
ESR1	rs9340799 6:151842246 · A/G	AA	chip	—	weak	ESR1 XbaI: see report Muscle, Bone and Sarcopenia. Information.

Thyroid

TSH is very hereditary (PDE8B, FOXE1, CAPZB). The desiodases activate (DIO1, DIO2: T4→T3) or inactivate hormones in tissues. In long-lived families, TSH tends to be higher and the lowest free T3.

Ageing mechanisms: Thyroid alterations; Decline of Thermogenesis; Metabolic Homeostasis Decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
TSH (60 loci)	No judgement (informative)	the 19th percentile · z=-0.84	Higher score = TSH. No judgment: the high-normal TSH accompanies family longevity; the low, more atrial fibrillation and bone loss. PGS004906 (PGS Catalog; Genome-wide signifier SNPs; Thyroid stimulating hormone concentration; PGP000639, PMID 37530217). Variants with genotype: 57 of 60 (covering of 94.9% coefficients).
Hypothyroidism (4535 variants)	Typical / intermediate	the 42nd percentile · z=-0.18	Higher score : higher risk. PGS000965 (PGS Catalog; penalized regression; Hypothyroidism/myxoedema; PGP000244, PMID 35324888). Variants with genotype: 4043 of 4524 (covering of 91.5% coefficients).
Hyperthyroidism (221 variants)	Typical / intermediate	the 56th percentile · z=0.09	Higher score : higher risk. PGS001042 (PGS Catalog; penalized regression; Thyrotoxicosis [hyperthyroidism] (time-to-event); PGP000244, PMID 35324888). Variants with genotype: 198 of 221 (covering of 91.0% coefficients).
Free T4 (loci of GWAS)	Typical / intermediate	the 28th percentile · z=-0.58	Higher score = lower free T4. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 72 of 75 loci with genotype. Studies: Choe EK (Free T4 levels); Rand SA (Thyroxine levels); Surendran P (Thyroxine levels); Teumer A (Free thyroxine concentration); Wei Y (Free thyroxine levels within normal range in pregnancy). Variants with genotype: 72 of 75 (covering of 94.0% coefficients).

Favourable

- DIO2 rs225014 TT

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DIO2	rs225014 14:80203237 · T/C	TT	chip	C (0; 34%)	moderate	Thr92Ala: C allele (92Ala) lowers the activity of DIO2 and T3 in tissues (brain, muscle); in treated hypothyroidism, some Ala/Ala carriers feel better with associated T3.

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DIO1	rs2235544 1:53909897 · C/A/G/T	CA	chip	A (1; 52%)	moderate	DIO1: C allele increases the activity (higher free T3, lower free T4); A = minor conversion.
FOXE1	rs965513 9:97793827 · A/C/G/T	GA	chip	A (1; 34%)	strong	FOXE1 (9q22): allele A increases hypothyroidism and thyroid cancer.
CAPZB	rs10458381 1:19442749 · C/A/T	CT	imputation DR2=1.00	T (1; 34%)	moderate	CAPZB: locus of TSH. Meaning: GWAS Catalog: Thyroid stimulating hormone levels (Williams AT 2023; p=2e-135); T ↑ allele

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PDE8B	rs4704397 5:77222617 · G/A/C	GG	chip	A (0; 40%)	strong	PDE8B: the strongest TSH locus (the highest TSH allele has a small effect, without disease). It does not have the highest TSH allele (without judgment). Meaning: GWAS Catalog: Thyroid stimulating hormone levels (Arnaud-Lopez L 2008; p=2e-20); allele A ↑
TSHR	rs1991517 14:81144239 · G/C/T	CC	chip	—	weak	TSHR D727E: associated with slightly lower TSH. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NR3C2	— — · —	—	SNP not found in the data	—	moderate	NR3C2: locus of TSH. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Cortisol and stress axis (HPA)

FKBP5 regulates the sensitivity of the glucocorticoid receptor (NR3C1); the T allele of FKBP5 rs1360780 prolongs the stress response. SERPINA6 (CBG) transports cortisol and is the main locus of morning cortisol.

Ageing mechanisms: Endocrine dysregulation; Chronic Inflammation; Sleep disturbance; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Cortisol (GWAS loci)	Typical / intermediate	the 33rd percentile · $z=-0.38$	Higher score = higher cortisol. Very few loci. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; $\ln(OR)$ or beta coefficients): 6 of 6 loci with genotype. Studies: Neumann A (Cortisol levels) Pott J (Cortisol levels) Surendran P (Cortisol levels). Variants with genotype: 6 of 6 (covering of 100.0% coefficients).

Favourable

- FKBP5 rs1360780 CC

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FKBP5	rs1360780 6:35639794 · T/A/C	CC	chip	T (0; 31%)	moderate	FKBP5 (2 intron): T allele increases the induction of FKBP5 by cortisol, makes the glucocorticoid receptor less sensitive and prolongs the stress response; interacts with stress in childhood (depression, PTSD).

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NR3C1	rs6190 5:143400772 · C/G/T	CC	chip	C (2; 97%)	moderate	ER22/23EK: the rare allele (T in strand +; 23K) gives slight resistance to glucocorticoids (better metabolic profile, more longevity in a study); C = common.
SERPINA6	rs12589136 14:94327349 · G/A/T	GG	chip	T (0; 20%)	moderate	SERPINA6 (CBG): main locus of morning cortisol. Meaning: GWAS Catalog: Cortisol levels (Hysi PG 2022; $p=1e-13$); allele T ↑

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CRHR1	rs110402 17:45802681 · G/A/C	GG	chip	—	weak	CRHR1: interaction with early stress in depression. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NR3C1	rs6195 5:143399752 · T/A/C	—	SNP not found in the data (Imputed with $DR2=0.73 < 0.8$)	G (—; 0%)	weak	N363S: G allele increases sensitivity to glucocorticoids (more metabolic effect of cortisol).

GH/IGF-1 axis and prolactin

Less GH/IGF-1 signal prolongs life in worms, flies and rats, and less active IGF1R variants are more frequent in centenarians; in humans, high IGF-1 is associated with some cancers and low IGF-1 fragility (U curve). Prolactin has few known loci.

Ageing mechanisms: Deregulation of Nutritional Sensing; Endocrine dysregulation; Sarcopenia

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
IGF-1 (23 442 variants)	Typical / intermediate	the 34th percentile · z=-0.36	Higher score = IGF-1. For longevity, the lower side is the most favourable (with protected muscle strength). PGS000687 (PGS Catalog; penalized regression; IGF-1 [nmol/L]; PGP000128, PMID 33462484). Variants with genotype: 20829 of 23442 (covering of 90.0% coefficients).
Prolactin (GWAS loci)	Favourable	the 12th percentile · z=-1.35	Higher score = higher prolactin. Few loci. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 6 of 6 loci with genotype. Studies: Pietzner M (Prolactin receptor levels); Pietzner M (Prolactin-inductive protein levels); Pietzner M (Prolactin-release peptide levels); Surapaneni A (Prolactin receptor level (9224_20)); Western D (Prolactin receptor (analyte X9224.20) levels). Variants with genotype: 6 of 6 (covering of 100.0% coefficients).

Favourable

- Prolactin (GWAS loci) (the 12th percentile)

Unfavourable

None.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IGF1R	rs2229765 15:98934996 · G/A/T	AG	chip	G (1; 57%)	weak	IGF1R (16 exon): allele A was associated with lower free IGF-1 and was more frequent in centenarians.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IGF1	rs35767 12:102481791 · A/C/G/T	GG	chip	—	moderate	IGF1 promoter: associated with IGF-1 in the blood.
IGFBP3	rs2854744 7:45921476 · G/C/T	GT	chip	—	moderate	IGFBP3 -202A>C: the main locus of IGFBP-3.
DRD2	rs1800497 11:113400106 · G/A	GG	chip	—	weak	Taq1A (ANKK1/DRD2): the allele A1 (T) lowers the D2 receptors; associated with prolactin a little higher in some studies. Information.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Thyroid alterations	—	DIO2 TT	Hypothyroidism (4535 variants) (the 42nd percentile); Hyperthyroidism (221 variants) (the 56th percentile); Free T4 (GWAS loci) (the 28th percentile); DIO1 CA; FOXE1 GA; CAPZB CT
Metabolic Homeostasis Decline	—	DIO2 TT	Hypothyroidism (4535 variants) (the 42nd percentile); Hyperthyroidism (221 variants) (the 56th percentile); Free T4 (GWAS loci) (the 28th percentile); DIO1 CA; FOXE1 GA; CAPZB CT
Thermogenesis decline	—	DIO2 TT	Hypothyroidism (4535 variants) (the 42nd percentile); Hyperthyroidism (221 variants) (the 56th percentile); Free T4 (GWAS loci) (the 28th percentile); DIO1 CA; FOXE1 GA; CAPZB CT
Decline of the Estrogen Signal	—	—	—
Endocrine dysregulation	—	DHEA-S (GWAS loci) (the 11th percentile); SHBG AG; SHBG CC; FKBP5 CC; Prolactin (GWAS loci) (the 12th percentile)	SHBG (19 328 variants) (the 66th percentile); SHBG AG; FAM9B A; Cortisol (loci of GWAS) (percentile 33); NR3C1 CC; SERPINA6 GG; IGF-1 (23 442 variants) (the 34th percentile)
Nutrient Sensing Deregulation	—	Prolactin (GWAS loci) (the 12th percentile)	IGF-1 (23 442 variants) (the 34th percentile)
Chronic Inflammation	—	FKBP5 CC	Cortisol (loci of GWAS) (percentile 33); NR3C1 CC; SERPINA6 GG
Neuroinflammation	—	FKBP5 CC	Cortisol (loci of GWAS) (percentile 33); NR3C1 CC; SERPINA6 GG
Osteoporosis	—	—	—
Thin Mass Loss	—	DHEA-S (GWAS loci) (the 11th percentile); SHBG AG; SHBG CC	SHBG (19 328 variants) (the 66th percentile); SHBG AG; FAM9B A
Sleep disturbance	—	FKBP5 CC	Cortisol (loci of GWAS) (percentile 33); NR3C1 CC; SERPINA6 GG
Insulin resistance	—	DHEA-S (GWAS loci) (the 11th percentile); SHBG AG; SHBG CC	SHBG (19 328 variants) (the 66th percentile); SHBG AG; FAM9B A
Sarcopenia	—	DHEA-S (GWAS loci) (the 11th percentile); SHBG AG; SHBG CC; Prolactin (GWAS loci) (the 12th percentile)	SHBG (19 328 variants) (the 66th percentile); SHBG AG; FAM9B A; IGF-1 (23 442 variants) (the 34th percentile)

Practical part

Ideas for the hormonal axis, crossed with the genetic plan v2 and current medication (isotretinoin, rapamycin, lithium 5 mg, tadalafil, dapagliflozin, retatrutide, telmisartan, LDN, melatonin). The current stack remains as it is; everything is suggestion (not applied). The genes do not indicate any hormone to replace; what changes the most is sleep, caloric deficit of retatrutide, training and medication.

1. Annual hormonal panel in the morning, fasting

Genetics: Testosterone p58; SHBG p66; estradiol p13; TSH p19

Total testosterone (7–10h), SHBG (to calculate free), estradiol by sensitive method, LH, prolactin, TSH, free T4 and IGF-1, 1×/year and always after large changes (retatrutide, finasteride). Compare with the previous values, not only with the reference range.

V2 genetic plan and medication: New; uses the harvests it already makes. • suggestion (not applied)

2. Do not lower estradiol

Genetics: Low genetic estradiol (p13); CYP19A1

In humans, estradiol protects bone, vessels and libido. No aromatase inhibitors or estrogen blockers (including supplements with chrysin or high dose DIM), except prescription.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

3. Testosterone: protecting with sleep, training and protein

Genetics: Typical testosterone (p58); SHBG p66

Sleep 7–9 h (one week at 5 h low testosterone 10–15%), strength training 3×/week, protein ~140 g/day and zinc + copper (already in the stack). A large and prolonged caloric deficit with retatrutide can decrease testosterone and T3: if they fall, review the deficit before thinking about hormones. Testosterone therapy only if there are repeated low values and symptoms (and suppress fertility).

V2 genetic plan and medication: Coherent with the plan (protein, creatine, zinc). • suggestion (not applied)

4. Finasteride and hormones (if used for hair)

Genetics: SRD5A2 rs523349 CC; see report

The finasteride low to DHT ~70% and slightly rises testosterone and estradiol; lower the PSA to half. If started, repeat the panel at 3 months.

V2 genetic plan and medication: See Hair (suggestion, not applied). • suggestion (not applied)

5. Thyroid: TSH with lithium

Genetics: Low genetic TSH (p19); DIO2 Thr/Thr; FOXE1 GA

Lithium can rise to TSH (rare with 5 mg): TSH and T4 free 1/year. Iodine and selenium already come from the stack minerals (50 µg and 55 µg); do not add algae or high dose iodine. If one day requires levothyroxine, with DIO2 Thr/Thr the T4 alone should be enough.

V2 genetic plan and medication: Compatible with the plan (S12 minerals). • suggestion (not applied)

6. Stress axis

Genetics: FKBP5 CC (favourable); typical cortisol (p33)

No sign of exaggerated response to stress. Keep anchors that lower the night cortisol: morning light, fixed waking time, early exercise. Low dose melatonin (S30 plan) does not move in cortisol.

V2 genetic plan and medication: Coherent with the v2 plan. • suggestion (not applied)

7. IGF-1: not trying to climb

Genetic IGF-1 p34; IGF1R AG

For longevity, IGF-1 in the middle-low third is the target; rapamycin already stops the mTOR downstream pathway. No GH secretagogues (MK-677, GH peptides) or colostrum "for IGF-1". The protein of ~140 g and creatine are for the muscle; if IGF-1 in the analyzes rises to the upper third, review the protein at night.

V2 genetic plan and medication: Coherent with the plan (rapamycin; protein ~140 g). • suggestion (not applied)

8. DHEA: no supplementary need

Genetics: High genetic DHEA-S (p11 in low DHEA-S score)

No genetic reason for DHEA; if DHEA-S is normal, supplementation only adds androgens and estrogens (act with the final isotretinoin, prostate).

V2 genetic plan and medication: Compatible with the plan. · suggestion (not applied)

Avoiding or with caution

- Inhibitors of aromatase and "anti-estrogen" supplements (already low genetic estradiol). suggestion (not applied)
- GH secretagogues and supplements to rise to IGF-1. suggestion (not applied)
- Testosterone or prohormones without repeated low values and medical follow-up. suggestion (not applied)
- High dose Iodine or algae in capsules (thyroid, lithium). suggestion (not applied)
- Very large caloric deficit during months (low testosterone and T3). suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. The CAG repeat of the androgen receptor and the 3 deletion of GHR are not read.
- Hormonal scores predict the genetic mean value; current medication (isotretinoin, retatrutide, dapagliflozin, telmisartan, tadalafil, lithium, LDN, melatonin) and sleep move more in the analysis than these genes.
- Lithium can decrease thyroid function (rare with 5 mg); isotretinoin can slightly lower free testosterone in some studies.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001988 (PGS Catalog; Penalized regression; Testosterone (male only); PGP000263, PMID 34995502 · PGS000694 (PGS Catalog; penalized regression; SHBG [nmol/L]; PGP000128, PMID 33462484 · PGS002916 (PGS Catalog; statistical method; Estradiol; PGP000393, PMID 36152628 · PGS004906 (PGS Catalog; Genome-wide signifier SNPs; Thyroid stimulating hormone concentration; PGP000639, PMID 37530217 · PGS000965 (PGS Catalog; penalized regression; Hypothyroidism/myxoedema; PGP000244, PMID 35324888 · PGS001042 (PGS Catalog; penalized regression; Thyrotoxicosis [hyperthyroidism] (time-to-event); PGP000244, PMID 35324888 · PGS000687 (PGS Catalog; penalized regression; IGF-1 [nmol/L]; PGP000128, PMID 33462484).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

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